

QFT 1

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1 Introduction

1.1 lecture 1.1

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Why study Quantum Field Theory?

QM and SR imply that particle number is not conserved. Taking a box full of particles, QM tells us that there is some uncertainty on the energy: $\Delta E \geq \frac{\hbar c}{L}$. SR tells us that $\Delta E = 2mc^2$. If we reduce the dimensions of the box enough, we can create a particle - antiparticle pair $\Rightarrow$ the number of particles is not conserved.

For this course we will consider $\hbar = c = 1$.

1.1.1 Classical Field Theory

$$\phi(t, \vec{x}) = \phi(x)$$

a field can be thought as a generalization of a collection of generalized coordinates in classical mechanics ($q_a(t) \Rightarrow \infty$ degrees of freedom).

We start by considering the action:

$$S = \int dt L(q_a, \dot{q}_a) = \int dt \underbrace{\int d^3x \mathcal{L}(\phi(x), \partial_\mu \phi(x))}_{L(t)}.$$

where $\mathcal{L}$ is the Lagrangian density.

One example of field is the vector potential.

Scalar Fields

$$\phi(x) \Rightarrow \phi'(x) = \phi(\Lambda^{-1}x)^1$$

¹A scalar field is a function that assigns a single scalar value to each point in spacetime. It can be written as $\phi : R^{1,3} \rightarrow R$, or, more explicitly, $\phi = \phi(x^\mu) = \phi(t, x, y, z)$. Under Lorentz transformations, a scalar field transforms as a scalar, i.e. its value at a given spacetime event is invariant.

under $x^\mu \Rightarrow (x')^\mu = \Lambda^\mu_\nu x^\nu$, where $\Lambda^\mu_\sigma \eta^{\sigma\rho} \Lambda^\nu_\rho = \eta^{\mu\nu}$ and

$$\eta^{\mu\nu} = \eta_{\mu\nu} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Simplest theory: Klein-Gordon

$$\mathcal{L}_{KG} = \frac{1}{2} \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} m^2 \phi^2$$

What are the equations of motion?

In classical mechanics we have

$$\frac{\delta S}{\delta q_a(t)} = 0$$

which gives the following equation of motion

$$\frac{\partial L}{\partial q_a} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_a} = 0$$

while in classical field theory we have

$$\frac{\delta S}{\delta \phi_a} = 0$$

which gives the following equation of motion

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} = 0.$$

So, for the KG theory, we can compute the equation of motion

$$m^2 \phi + \partial_\mu \partial^\mu \phi = 0.$$

Noether's Theorem Every continuous symmetry of the action gives rise to a conserved current $j^\mu(x)$. That means $\partial_\mu j^\mu = 0$.

A conserved current implies a conserved charge. To see that we define

$$Q = \int_{\mathcal{R}^3} d^3x \frac{dj^0}{dt} = - \int_{\mathcal{R}^3} d^3x \vec{\nabla} \cdot \vec{j} = 0$$

assuming $j \rightarrow 0$ as $|\vec{x}| \rightarrow \infty$.

This tells us that charges also conserve locally

$$Q_V = \int_V d^3x j^0 \Rightarrow \frac{dQ_V}{dt} = - \int_A \vec{j} \cdot d\vec{S}.$$